

## SAFETY DATA SHEET

Revision Date 01-Apr-2024

Revision Number 4

### 1. Identification

**Product Name** Kinase buffer V (5X)  
**Cat No. :** J62045  
**Synonyms** No information available  
**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

#### Label Elements

##### **Hazard Statements**

May cause respiratory irritation



**Precautionary Statements****Hazards not otherwise classified (HNOC)**

None identified

**3. Composition/Information on Ingredients**

| Component                        | CAS No     | Weight % |
|----------------------------------|------------|----------|
| Water                            | 7732-18-5  | 96.44    |
| 4-Morpholinepropanesulfonic acid | 1132-61-2  | 2.09     |
| Manganous chloride tetrahydrate  | 13446-34-9 | 0.99     |
| Magnesium chloride               | 7786-30-3  | 0.48     |

**4. First-aid measures**

|                                            |                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                  |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                            |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                   |

**5. Fire-fighting measures**

|                                         |                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                             |
| <b>Flash Point</b>                      | No information available                                                                                                                                             |
| <b>Method -</b>                         | No information available                                                                                                                                             |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                             |
| <b>Explosion Limits</b>                 |                                                                                                                                                                      |
| <b>Upper</b>                            | No data available                                                                                                                                                    |
| <b>Lower</b>                            | No data available                                                                                                                                                    |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                             |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                             |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides. Hydrogen chloride. Metal oxides.**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**Health  
2Flammability  
0Instability  
0Physical hazards  
-**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal.

Up

**7. Handling and storage****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**8. Exposure controls / personal protection****Exposure Guidelines**

| Component                       | ACGIH TLV                                                 | OSHA PEL                                                               | NIOSH                                                                                | Mexico OEL (TWA)           |
|---------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------|
| Manganous chloride tetrahydrate | TWA: 0.02 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> | (Vacated) Ceiling: 5 mg/m <sup>3</sup><br>Ceiling: 5 mg/m <sup>3</sup> | IDLH: 500 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup><br>STEL: 3 mg/m <sup>3</sup> | TWA: 0.2 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

None under normal use conditions.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:**

Particle filter.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**9. Physical and chemical properties****Physical State**

Liquid

**Appearance**

No information available

**Odor**

No information available

**Odor Threshold**

No information available

**pH**

No information available

**Melting Point/Range**

No data available

|                                        |                          |
|----------------------------------------|--------------------------|
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                                                       |
|----------------------------------|-----------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                            |
| Stability                        | Stable under normal conditions.                                       |
| Conditions to Avoid              | Incompatible products.                                                |
| Incompatible Materials           | Strong oxidizing agents                                               |
| Hazardous Decomposition Products | Nitrogen oxides (NOx), Sulfur oxides, Hydrogen chloride, Metal oxides |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                              |
| Hazardous Reactions              | None under normal processing.                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component                        | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation |
|----------------------------------|---------------------------|---------------------------|-----------------|
| Water                            | -                         | -                         | -               |
| 4-Morpholinepropanesulfonic acid | LD50>2g/kg (rat)          | Not listed                | Not listed      |
| Manganous chloride tetrahydrate  | LD50 = 1484 mg/kg ( Rat ) | Not listed                | Not listed      |
| Magnesium chloride               | LD50 = 2800 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component            | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------|-----------|------------|------------|------------|------------|------------|
| Water                | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| 4-Morpholinepropanes | 1132-61-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                 |            |            |            |            |            |            |
|---------------------------------|------------|------------|------------|------------|------------|------------|
| ulfonic acid                    |            |            |            |            |            |            |
| Manganous chloride tetrahydrate | 13446-34-9 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Magnesium chloride              | 7786-30-3  | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component                        | Freshwater Algae    | Freshwater Fish                            | Microtox                                                                                                                                                | Water Flea           |
|----------------------------------|---------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 4-Morpholinepropanesulfonic acid | Not listed          | Not listed                                 | Not listed                                                                                                                                              | >100 mg/L 48h        |
| Manganous chloride tetrahydrate  | ErC50 = 61 mg/l     | Not listed                                 | Not listed                                                                                                                                              | Not listed           |
| Magnesium chloride               | EC50: 2200 mg/L/72h | Pimephales promelas:<br>EC50: 2.12 g/L/96H | EC50 Pseudomonas putida:<br>EC50: 26,14 g/L/h<br>Photobacterium phosphoreum: EC50: 36,3 mg/L/30 min<br>Photobacterium phosphoreum: EC50: 77,2 mg/L/24 h | EC50 : 1400 mg/L/24h |

**Persistence and Degradability** based on information available. May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component                        | log Pow |
|----------------------------------|---------|
| 4-Morpholinepropanesulfonic acid | -2.94   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

### United States of America Inventory

| Component                        | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------------|------------|------|-----------------------------------------------|-----------------------------|
| Water                            | 7732-18-5  | X    | ACTIVE                                        | -                           |
| 4-Morpholinepropanesulfonic acid | 1132-61-2  | X    | ACTIVE                                        | -                           |
| Manganous chloride tetrahydrate  | 13446-34-9 | -    | -                                             | -                           |
| Magnesium chloride               | 7786-30-3  | X    | ACTIVE                                        | -                           |

#### Legend:

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)** Not applicable

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                        | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL        |
|----------------------------------|------------|-----|------|-----------|-------|------|------|------|-------|-------------|
| Water                            | 7732-18-5  | X   | -    | 231-791-2 | X     | X    |      | X    | X     | KE-35400    |
| 4-Morpholinepropanesulfonic acid | 1132-61-2  | X   | -    | 214-478-5 | -     | -    | X    | X    | X     | 2010-3-4223 |
| Manganous chloride tetrahydrate  | 13446-34-9 | -   | -    | -         | X     | -    |      | X    | X     | -           |
| Magnesium chloride               | 7786-30-3  | X   | -    | 232-094-6 | X     | X    | X    | X    | X     | KE-22691    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component                       | CAS No     | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|---------------------------------|------------|----------|-------------------------------|---------------------------------|
| Manganous chloride tetrahydrate | 13446-34-9 | 0.99     | 1.0 %                         | -                               |

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

#### Clean Air Act

| Component                       | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|---------------------------------|-----------|-------------------------|-------------------------|
| Manganous chloride tetrahydrate | X         |                         | -                       |

**OSHA** - Occupational Safety and Health Administration Not applicable

#### CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive

Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                       | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------------------|---------------|------------|--------------|----------|--------------|
| Water                           | -             | -          | X            | -        | -            |
| Manganous chloride tetrahydrate | -             | X          | X            | X        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
 DOT Marine Pollutant N  
 DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

No information available

**Authorisation/Restrictions according to EU REACH**

Not applicable

| Component                        | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water                            | 7732-18-5  | -                                                                   | -                                                                             | -                                                                                                     |
| 4-Morpholinepropanesulfonic acid | 1132-61-2  | -                                                                   | -                                                                             | -                                                                                                     |
| Manganous chloride tetrahydrate  | 13446-34-9 | -                                                                   | -                                                                             | -                                                                                                     |
| Magnesium chloride               | 7786-30-3  | -                                                                   | -                                                                             | -                                                                                                     |

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                        | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Water                            | 7732-18-5  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| 4-Morpholinepropanesulfonic acid | 1132-61-2  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Manganous chloride tetrahydrate  | 13446-34-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Magnesium chloride               | 7786-30-3  | Listed         | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component | CAS No | Seveso III Directive | Seveso III Directive | Rotterdam | Basel Convention |
|-----------|--------|----------------------|----------------------|-----------|------------------|
|-----------|--------|----------------------|----------------------|-----------|------------------|

|                                     |            | (2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | (2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Convention (PIC) | (Hazardous Waste) |
|-------------------------------------|------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|-------------------|
| Water                               | 7732-18-5  | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |
| 4-Morpholinepropanesulfonic<br>acid | 1132-61-2  | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |
| Manganous chloride<br>tetrahydrate  | 13446-34-9 | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |
| Magnesium chloride                  | 7786-30-3  | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
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www.thermofisher.com

**Revision Date** 01-Apr-2024  
**Print Date** 01-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**